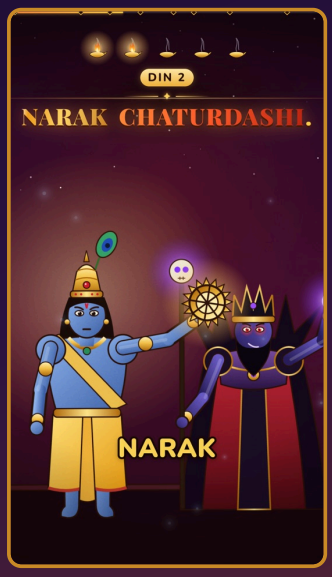
A CLAUDE + HYPERFRAMES PLAYBOOK

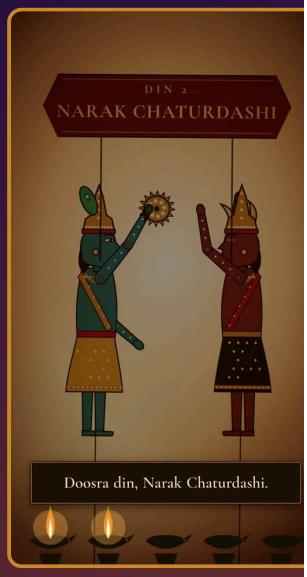
2D Mythology Reels, drawn entirely in code

This playbook takes you from a story idea to a finished 20–60 second 9:16 reel. Claude draws every character, set and prop as SVG and animates it with GSAP. No AI image or video generators are used.









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Contents

1. The idea and the ground rules
2. One-time setup
3. Step 1: Write the script
4. Step 2: Voiceover and word timings
5. Step 3: Start the project (fork the kit)
6. Step 4: Characters, the drawn kit
7. Step 5: Scenes and frame layout
8. Step 6: Animation
9. Step 7: Shots, titles and captions (build.py)
10. Step 8: Check, fix, render
11. Two styles: Makardhwaj and Shadow Puppet
12. Troubleshooting
13. Working with Claude: prompts that work
14. Worked example: Sanjeevni (20 s)

> **Quick path.** Once setup is done, the whole process is one sentence to Claude: "*Use mythology-kit-reel to make a 20 sec reel of <story>. Script first.*" The rest of this guide explains what happens behind that sentence, so you can steer it, fix it, or do it by hand.

1 The idea and the ground rules

Each reel is an **HTML page that renders to video**. Python scripts write SVG drawings of every scene, GSAP animates them on a single timeline, and HyperFrames records the page frame by frame into an MP4 with your voiceover.

Why code-drawn instead of AI-generated

- **Consistency:** Hanuman has the same face, crown and gada in every shot, because he comes from the same function.
- **Control:** motion lands exactly on a word ("utha *laaye*!"), not where a video model happens to put it.
- **Cost and reuse:** there are no credits to buy, and every character you draw becomes part of the kit for the next reel.

> **Hard rule.** Never use Higgsfield, Arcads, Pixa, Runway or any other AI image or video tool for the art. Claude writes every picture as SVG. If someone suggests "just generate it", say no.

The format

Setting	Value
Size	1080 × 1920 (9:16), 30 fps
Length	20 s (6 scenes) up to 60 s (12–16 scenes)
Audio	Voiceover (48 kHz WAV) plus synthesized SFX: whoosh on cuts, boom on hits, shimmer on reveals
Text	A big title per scene (top) and word-by-word captions with the key words in gold (lower third)
File	MP4 under 30 MB for posting

2 One-time setup

Install (macOS)

```
brew install node ffmpeg python
# Google Chrome must be installed (HyperFrames renders with it)
npx --yes hyperframes@0.8.214 --version      # pinned version used by every project
brew install whisper-cpp                     # optional: word timings for your voiceover
```

Folder layout

```
~/Instagram/
  rudra-hanuman/art.py          Hanuman, Ravana, Vibhishana, limb/eyes/mouth/crown helpers
  makardhwaj-short/drawn.py     Ram, Lakshman, Makardhwaj, Ahiravan, sets, animation helpers
  madhubani-reel/version-a/     the template project you fork (dscenes.py + mscenes.py)
  madhubani-reel/version-b/     the shadow-puppet style (single build.py)
  premium.py                    caption styling shared by all reels
  <new-reel>/                  each new video gets its own folder
```

Claude skills that drive this

Skill	Use it for
`mythology-kit-reel`	The fast path: forks the kit and builds a reel in the Makardhwaj style
`2d-explainer-reel`	The general process for any 2D explainer, including shortening a long VO
`hyperframes` , `hyperframes-cli`	The composition rules, lint, snapshot and render

3 Step 1: Write the script

Write the script first and get it approved before any drawing. The drawing work follows from the script.

The 20-second formula

Beat	Time	Job
Hook	0–3 s	Drop into the crisis. No "aaj hum dekhenge". Example: "Shakti baan laga... Lakshman ji behosh."
Stakes	3–6 s	Add a clock or a cost. Example: "suraj ugne se pehle"
Action	6–9 s	The hero moves, with one big visual verb (leap, lift, strike)
Twist	9–13 s	A problem inside the problem. Example: "kaun si booti?"
Payoff	13–16 s	The iconic image people know (the whole mountain on one palm)
Close	16–20 s	Resolution plus a chant or moral the viewer repeats: "Bolo, Jai Shri Ram!"

- About **55 Hinglish words per 20 s** (roughly 140 words for 55 s).
- **One action per scene.** If a sentence has two verbs, split it into two scenes.
- Pick a **through-line visual** that changes across the video: five diyas lighting one per day, a sun-timer draining. It ties the cuts together and rewards re-watching.
- Every scene gets a 1–3 word **title** in caps, which is also the thumbnail text.

Template (SCRIPT.md)

```
| # | Time  | Voiceover (Hinglish)                    | Visual + motion                       |
Title          |
|---|-------|-----------------------------------------|---------------------------------------|---
-------------|
| 1 | 0–3   | Shakti baan laga... Lakshman ji behosh.   | Night camp, Lakshman lying, Ram kneels |
SHAKTI BAAN    |
| 2 | 3–6   | ...                                     | ...                                   |
...            |
```

4 Step 2: Voiceover and word timings

Record

- Record in one take in a quiet room, with the phone about 20 cm away. Leave a half-second pause between beats so cuts land cleanly.
- Speak slightly faster than normal. Reels reward energy.
- Convert and trim to the exact length:

```
ffmpeg -i raw.m4a -ar 48000 -ac 1 -af "loudnorm=I=-16:TP=-1.5,afade=t=out:st=19.6:d=0.4" -t 20
assets/audio/voiceover.wav
```

Cutting from someone else's reel (only with rights)

List the keep-ranges and join them with tiny fades so the splices don't click:

```
ffmpeg -i src.wav -af "atrim=0:4.45,afade=t=in:d=0.03,afade=t=out:st=4.41:d=0.04" p1.wav   # one per
piece
ffmpeg -i p1.wav -i p2.wav ... -filter_complex concat=n=8:v=0:a=1 vo20.wav
```

Word timings: assets/words.txt

Use one line per sentence, with each word followed by its start time in seconds and separated by `|`. Captions, title pops and punch-ins all key off these times.

```
Shakti 0.00|baan 0.42|laga... 0.71|Lakshman 1.60|ji 2.05|behosh. 2.30
Vaidya 3.05|Sushen 3.40|bole 3.85|suraj 4.30|ugne 4.62|se 4.90|pehle 5.02|Sanjeevni 5.30|chahiye.
5.85
```

```
whisper-cli -m ggml-large-v3.bin -f assets/audio/voiceover.wav -ml 1 -oj   # word-level JSON, then
convert
```

> **Gotcha.** Whisper garbles audio that was spliced together. Transcribe the *original* take instead, then shift each word by its piece's offset.

5 Step 3: Start the project (fork the kit)

```
cd ~/Instagram
cp -R madhubani-reel/version-a sanjeevni && rm -rf sanjeevni/renders
cd sanjeevni
cp mscenes.py sscenes.py            # your scene file; keep the helpers, delete m01..m08
```

The new folder sits one level higher than version-a, so fix three paths:

File	Change
`build.py`	`sys.path.insert(0, "../..")` becomes `".."`, and `import mscenes as drawn` becomes `import sscenes as drawn`
`dscenes.py`	`../../makardhwaj-short` becomes `../makardhwaj-short`, and `../../rudra-hanuman` becomes `../rudra-hanuman`
`meta.json`, `index.html`	New id, and every duration set to TOTAL (20.0)

What each file does

File	Role
`sscenes.py`	Your story. A `sNN()` function draws each scene and an `aNN()` function animates it.
`dscenes.py`	Imports the kit and adds townsfolk, houses, palace, toran and fireworks
`build.py`	Shot list, titles and captions. Writes `compositions/*.html`.
`index.html`	Root timeline: scenes, titles, captions, voiceover and SFX tracks
`assets/`	`audio/voiceover.wav`, `audio/sfx.wav`, `words.txt`, `grain.png`

6 Step 4: Characters, the drawn kit

Every character is a Python function that returns an SVG `<g>` drawn around the feet at (0, 0), about 900 units tall. You place it with `place(svg, x, y_feet, scale, flip=False)`.

Character	Call	Poses / options
Hanuman	`hanuman(pid, expr, pose, rudra=False, prop="gada")`	stand, speak, dive (flying); prop gada / trishul / None
Ram, Lakshman	`ram(pid, expr, pose)`, `lakshman(...)`	prince body with bow
Ravana, Vibhishana	`ravana(pid, expr, pose)`	10 heads; point / speak
Makardhwaj, Ahiravan	`mk.makardhwaj(...)`, `mk.ahiravan(...)`	guard, staff, cast
Deva / Krishna	`deva(pid, skin, sash, expr, pose)`, `krishna(...)`	stand, bless, lift, kalash, chakra
Lakshmi, Sita, villagers	`lady(...)`, `sita`, `man(hair="pagdi")`, `kid`, `mom`	stand, diya, wave
Family of four	`family(tag, y, pose, expr)` plus `family_js`	fills an empty frame instantly

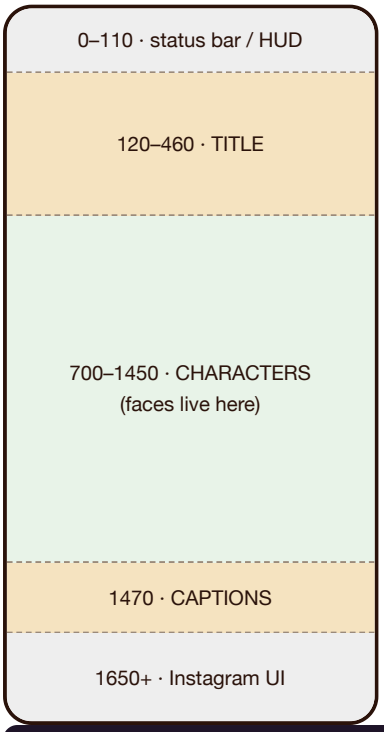
> **Expressions that exist:** calm · closed · awe · sneer · fierce · strain · shock · worried. Anything else (for example "happy") crashes the build. Show joy with `calm`, an open pose and warm light.

Drawing a new character (same idiom)

```
def sushen(pid, expr="calm"):
    o = f'<g id="{pid}">' + man(pid + "b", "gSkinB", "gCream", "gCream", expr, "pagdi", "stand")
    o += f'<path d="M-50,-640 Q0,-520 50,-640 Q0,-600 -50,-640 Z" fill="#f4f1ea" stroke="{INK}"
stroke-width="4"/>'  # beard
    o += f'<path d="M150,-700 L170,0" stroke="#6a3a1a" stroke-width="14" stroke-linecap="round"/>'
# staff
    return o + '</g>'
```

- Use an ink stroke 4–6 wide, two-stop gradient fills from `MDEFS`, joints made with `limb()`, and eyes and mouth from `eyes()` / `mouth()`.
- Before using a new character in scenes, render it alone on a sheet and look at it.

7 Step 5: Scenes and frame layout



Every scene function returns a full 1080 × 1920 SVG, built back to front:

1. **Sky**: `sky(id, stops)`, plus `stars` and `clouds`
2. **Far set**: `houses`, `palace`, `lanka`, `sea_dusk`, mountains
3. **Light**: `glow(x, y, r)` behind the hero, and `rays`
4. **Mid-ground fill**: lanterns, herbs, soldiers, rain. **Never leave this empty.**
5. **Characters**, each wrapped in `mv(id, ...)` so you can move it
6. **Front**: `floor`, diya row, rangoli, `petals`, `motes`

```
def s04():
    o = sky("b4", [(0, "#0a1830"), (.6, "#3a5a8a"), (1, "#c8dcf0")]) + stars(4, 60, 600)
    o += dronagiri(540, 1500)                                    # far set + herb glows
    o += mv("hn4", place(hanuman("h4", "worried", "stand"), 540, 1450, .85))
    o += mv("q4", f'<text x="540" y="560" font-size="160" text-anchor="middle" fill="#ffd27a">?
</text>')
    return o + motes(4, 50, "#e0f0ff", 300, 1700, 4)
```

Lesson from the Diwali reel: the first cut had scenes with only a title on a bare sky, and they read as "empty". The fix was a family of four plus hanging lanterns. Every scene needs a character in the middle band and something in the mid-ground.



Version A frames: the intro with the family and lanterns, Krishna against Narakasur, Govardhan, and the finale.

8 Step 6: Animation

Each `aNN(t0, t1)` returns a list of GSAP lines placed at **absolute seconds**. Use the times in words.txt so actions hit their words.

Helper	What it does	Example
`tw(sel, from, to, t, dur, ease)`	One fromTo tween	`tw("#hn3", "x:-600", "x:1400", 6.6, 1.8, "power2.in")`
`yoyo(sel, from, to, t0, t1, period, origin)`	Repeating swing or float	lantern sway, mountain wobble
`breathe(pid, t0, t1)`	Subtle chest rise, so characters never freeze	every character, every scene
`blink(sid, t0, t1)`	Blinks every eye in the scene	every scene with faces
`talk(pid, a, b)`	Lip-sync between two times	Sushen's line
`flicker(sid, t0, t1)`	Flames dance	diyas, torches
`pop(sel, t, origin)`	Elastic pop-in	"?", tilak, lit diya

Motion rules

- **Something always moves:** breathe and blink are on every character, motes drift, and the camera pushes slowly.
- **One hero action per scene**, landing on its key word.
- **Hits:** a white flash of 0.15 s, a 6-frame camera shake, and a boom in the SFX.
- Rotations of SVG groups need `transformOrigin` or `svgOrigin`. Flames in an SVG `<g>` should only fade, never scale.
- Deterministic only: use `random.Random(seed)` in Python and never `Math.random` in the page.

```
def a05(t0, t1):
    return breathe("h5", t0, t1) + blink("s05", t0, t1) + [
        tw("#mt5", "y:300", "y:0", 13.3, .9, "back.out(1.4)"),            # mountain rises on "utha"
        yoyo("#mt5", "rotation:-1", "rotation:1", 14.2, t1, .8, origin="50% 100%"),
        tw("#hn5", "x:0,y:0", "x:-700,y:-900", 15.2, .8, "power2.in")]      # fly out
```

9 Step 7: Shots, titles and captions (build.py)

```
TOTAL = 20.0
XF = 0.4                        # crossfade between shots
# id, image, start, camera [s0, x0, y0, s1, x1, y1], mote colour, punch-ins (abs s)
SHOTS = [
    ("s01", "", 0.0,  [1.14, 0, 40, 1.04, 0, 0],   "#ffd27a", [1.6]),
    ("s02", "", 3.0,  [1.04, 0, 40, 1.14, 0, -40], "#fff0b8", [5.3]),
    ...
]
# id, start, end, kicker, big, sub
TITLES = [("t01", 0.2, 2.9, "", "SHAKTI BAAN", ""), ("t02", 3.1, 5.9, "", "SURYODAY SE PEHLE", ""),
...]
premium.KEY |= {"shakti", "sanjeevni", "himalaya", "parvat", "ram!"}   # caption words shown in gold
```

- **Camera:** alternate push-in and pull-out between scenes, and keep scale between 1.04 and 1.16.
- **Punch-ins** are quick 6 % zooms. Put them on the strongest word of each scene.
- **Captions** show 2–3 words per group with the active word gold, at top 1470 px.
- **SFX** is generated once into `assets/audio/sfx.wav`. Re-run `python3 build.py --sfx` after changing cuts.

10 Step 8: Check, fix, render

```
python3 build.py
npx --yes hyperframes@0.8.214 lint                      # 0 errors before anything else
npx --yes hyperframes@0.8.214 snapshot --at 1.5,4.5,7.5,11,14.5,18.5
#  open .hyperframes/artifacts/snapshots/<id>/contact-sheet.jpg
npx --yes hyperframes@0.8.214 render -o renders/Sanjeevni_1080x1920.mp4 --quiet
```

Contact-sheet checklist (look at every frame)

Check	Fix if not
Is anything happening in the middle band?	Add a character, family, lanterns, herbs or clouds
Are the faces clear of the title and captions?	Move feet to y 1400–1480, or scale to 0.7–0.9
Is anything cut off at the edges?	Pull x in from the edges by 80 px or more
Can you read the title against the sky?	Darken the sky top or add a glow
Does the action land on its word?	Retime the `tw` to the words.txt time
Does the end frame feel final?	Flash plus rays plus petals, held for at least 1.5 s

Finally, **watch the whole video once with sound**, because frames alone miss sync problems. Keep the file under 30 MB. If it's bigger, re-encode:

```
ffmpeg -i in.mp4 -c:v libx264 -crf 22 -preset slow -c:a aac -b:a 160k out.mp4
```

11 Two styles: Makardhwaj and Shadow Puppet

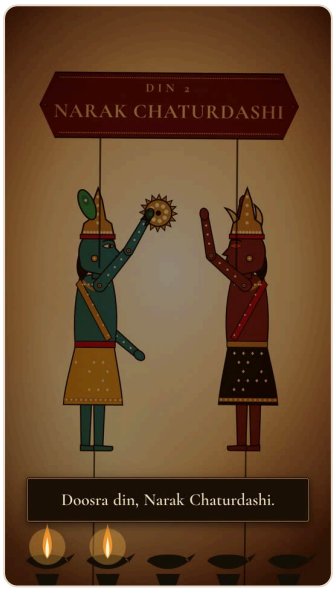
A · Makardhwaj kit (default)

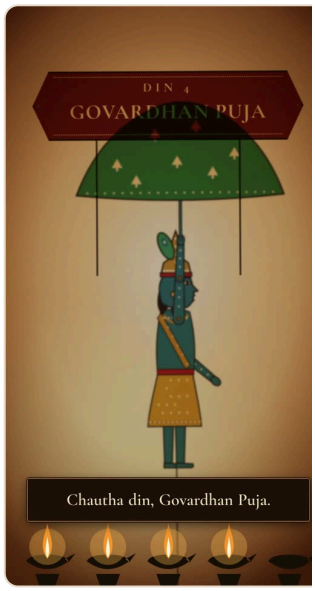




- Ink-outlined gradient characters with jointed limbs and big expressive eyes
- Deep indigo nights, saffron and gold light, glows and motes
- Best for action, dialogue and stories with many characters
- Template: `madhubani-reel/version-a`

B · Shadow puppet (tholu bommalata)





- Dyed-hide puppets in profile, with carved holes, rivets and rods
- A cotton screen lit by oil lamps from behind. Puppets press onto the screen (blur goes to sharp) and lift off it.
- Best for folk tales, festivals, and "light beats darkness" stories
- Template: `madhubani-reel/version-b/build.py` (a single file)

> **Picking a style.** If the story is about light (Diwali, Amavasya, dawn), the shadow puppet look turns the theme into the lighting itself, because each lamp brightens the screen. If it's about a hero doing something (Sanjeevni, Lanka dahan), Makardhwaj reads better because of its faces and action poses.

12 Troubleshooting

Symptom	Cause	Fix
`KeyError: 'happy'`	The expression doesn't exist	Use one of the 8 listed expressions
Lint: `gsap_fullscreen_overlay_starts_visible`	A dark wash or flash starts visible	CSS `opacity:0` plus `tl.fromTo`
Flames drift away from their lamps	Scaling an SVG `<g>` with transformOrigin	Fade the opacity only
Captions cover faces	Characters placed too low or too big	Feet at 1400–1480, scale 0.7–0.9
A prop hidden behind a body	Draw order	Draw it after the character, or raise its y
A scene stays dark	The overlay never fades out	Add a `tw("#dk", "opacity:.7", "opacity:0", ...)`
ffmpeg SFX mix hangs	An `amix` with a long input list	Kill it (`pkill ffmpeg`). Generate SFX only when missing or with `--sfx` . macOS has no `timeout` command.
Captions out of sync	Whisper read spliced audio	Use timings from the original take, shifted by the cut offsets
Render is over 30 MB	Grain and motes at high bitrate	Re-encode at `-crf 22`

13 Working with Claude: prompts that work

Goal	Say this
New reel	"Use mythology-kit-reel to make a 20 sec reel of Lanka Dahan. Script first."
From someone's reel	"Use the VO from <link>, trim it to 20 sec, and make it in the Makardhwaj style. I have rights."
Two styles	"Make version A in the current kit and version B in the best style from <repo>."
Fix a frame	Send a screenshot with one line: "this looks very empty", "the face is under the caption".
Swap the voice	"Here is my recording. Re-time the reel to it."
Change one scene	"In scene 4, make Hanuman look left and right twice before the question mark."

Approval points. Claude stops for your OK at the script, at the character sheet for new characters, and at the contact sheet before the final render. Answering quickly keeps a 20 s reel to one session.

Rights. Only reuse another creator's voiceover when you have permission. Otherwise record your own or use TTS.

14 Worked example: Sanjeevni (20 s)

Through-line: a sun-timer ring in the top-right drains from 3 s to 16 s, and the sun rises out of it at 16 s. **Palette:** night indigo, then snow blue, then dawn saffron.

#	Time	VO	Visual and motion	Title
s01	0–3	Shakti baan laga... Lakshman ji behosh.	Night camp. Lakshman lying (rotated −90°), arrow beside him, Ram kneeling, Hanuman worried. Torches flicker, slow push-in.	SHAKTI BAAN
s02	3–6	Vaidya Sushen bole: suraj ugne se pehle Sanjeevni chahiye.	Sushen (villager in a pagdi, with beard and staff) talks and points north. The timer ring pops in.	SURYODAY SE PEHLE
s03	6–9	Hanuman ji ek chhalaang mein Himalaya ki or ude.	Hanuman grows from 0.6 to 1.3 and crouches, then flies in the dive pose. Sea and clouds stream past.	JAI BAJRANGBALI
s04	9–13	Par pahaad pe hazaaron bootiyan chamak rahi thi... kaun si?	Snowy Dronagiri with about 40 glowing herbs flickering. Hanuman, worried, turns his head left and right, and a "?" pops in.	DRONAGIRI
s05	13–16	To Hanuman ji poora parvat hi utha laaye!	New lift pose with the mountain on his palm. Rocks fall, he flies out top-left, and the timer glows red.	POORA PARVAT
s06	16–20	Suraj se pehle Lakshman ji uthe. Bolo, Jai Shri Ram!	Dawn with rays and the sun rising. Lakshman sits up, Ram and Hanuman embrace, petals fall, and a flash lands on "Jai Shri Ram".	JAI SHRI RAM

New drawing work

- Sushen (beard and staff over `man`), the arrow, `dronagiri()` with herb glows, a Hanuman **lift** pose (copy the arm angles from the `deva` "lift" pose), and the sun-timer HUD (copy `days_html`).

SFX and gold words

- Whoosh on every cut, a low boom at 6.5 s (leap) and 13.3 s (lift), and a shimmer at 16.2 s (dawn).
- `premium.KEY`: shakti, sanjeevni, himalaya, bootiyan, parvat, lakshman, ram!

> **Your next move:** record the six lines in one take, save the file as `voiceover.wav`, and tell Claude: "*Use mythology-kit-reel to make the Sanjeevni reel. Here's my voiceover.*"